This system performs a **digital eye test for children** using camera, computer vision, and AI.
Since children may sometimes **guess answers or intentionally give wrong responses**, the system also checks **behavior patterns** to determine if the test results are reliable.

---

1. Detecting Intentional Wrong Answers

The system **cannot directly know a child's intention**, but it can detect patterns that suggest:

- guessing
- not paying attention
- intentionally giving wrong answers

To do this, we check **response behavior and eye tracking data**.

---

2. Repeat Question Check

Sometimes the system shows the **same letter again later in the test**.

Example:

Round	Letter	Child Answer
1	E	P
4	E	E

This means the child **can see the letter**, but the first answer was wrong.

So the system **reduces the reliability score**.

---

3. Response Time Analysis

Reading a letter normally takes **around 1–2 seconds**.

Time	Meaning
< 0.4 sec	Probably guessing
0.8 – 3 sec	Normal reading
5 sec | Difficulty seeing |	

If the child answers **too quickly many times**, it may be guessing.

---

4. Difficulty Pattern Check

Vision normally follows this pattern:

- **Large letters → easy**

- **Medium letters → moderate**
- **Small letters → difficult**

Example of a normal result:

Large letters → correct
Medium letters → some mistakes
Small letters → more mistakes

Suspicious pattern:

Letter Size	Result
Large	Wrong
Medium	Correct
Small	Correct

This is unlikely, so the system marks it as **unreliable behavior**.

---

5. Eye Tracking

The camera checks whether the child **actually looks at the letter**.

Technologies used:

- MediaPipe
- OpenCV

The system tracks:

- gaze direction
- how long the child looks at the letter
- blink rate

If the child **answers without looking at the letter**, it is probably guessing.

---

6. Attention Monitoring

The system also checks whether the child is **paying attention to the screen**.

It monitors:

- head movement
- eye direction
- blinking

If the child **looks away from the screen frequently**, the test reliability decreases.

---

7. Control Questions

The system sometimes shows **very easy letters**.

Example:

Large letter **O**

If the child answers this wrong, it likely means **they are not cooperating**.

---

8. Reliability Score

All behavior checks contribute to a **test reliability score**.

Example scoring system:

Start score = 100

- inconsistent answers → -10
- very fast guessing → -5
- not looking at screen → -10
- impossible difficulty pattern → -15
- wrong answer on easy letter → -20

Final reliability levels:

Score	Meaning
80 – 100	Reliable test
60 – 80	Moderate reliability
Below 60	Test unreliable

If reliability is too low, the system asks the user to **repeat the test**.

---

9. Machine Learning Behavior Analysis

Instead of analyzing only individual answers, the system can analyze **the entire test session**.

Each session creates a **behavior data record**.

Example features:

- accuracy
- average response time
- response time variation

- gaze focus score
- blink rate
- movement distance
- accuracy on large letters
- accuracy on small letters

This data is used to train a machine learning model using:

- Scikit-learn

The model can classify behavior as:

Behavior	Meaning
Cooperative	Child trying honestly
Guessing	Random answers
Not paying attention	Child distracted
Intentional wrong	Deliberate incorrect answers

---

10. Adaptive Vision Testing

Instead of showing fixed rows of letters, the system **changes difficulty automatically**.

Basic rule:

Correct answer → smaller letter (harder)
Wrong answer → bigger letter (easier)

Example:

Step	Letter Size	Result
1	40px	Correct
2	30px	Correct
3	20px	Wrong
4	30px	Correct
5	25px	Correct

The system slowly finds the **smallest readable letter size**.

---

11. Vision Score Estimation

The smallest readable letter size is used to estimate **visual acuity**.

Example mapping:

Letter Size	Estimated Vision
80px	20/200
60px	20/100
40px	20/60
30px	20/40
20px	20/20

Example result:

Estimated Vision = **20/40**

Important: this is **only a screening result**, not a medical diagnosis.

---

12. System Workflow

Complete system flow:

Camera Input
↓
Face Detection
↓
Distance Check
↓
Letter Display
↓
Eye Tracking
↓
Child Answer
↓
Behavior Analysis
↓
Adaptive Difficulty Adjustment
↓
Vision Score Calculation

Computer vision tools:

- OpenCV
- MediaPipe

Machine learning:

- Scikit-learn

---

13. Example Final Result

Example output:

Distance Vision Test Completed

Estimated Visual Acuity: **20/40**
Near Vision Score: **8/10**

Test Reliability: **82%**
Behavior Classification: **Cooperative**

Recommendation:
Visit an eye specialist if vision problems continue.